OPSINTEL AI

# Evidence-based pipeline investigation

From failed overnight run to traceable next action - powered by Exasol Personal

EXATHON INDIA 2026 / TRACK 04: THE WORK BEHIND THE WORK

Public prototype • MIT licensed • Synthetic data

THE PROBLEM

# The failure is visible. The explanation is scattered.

MANUAL

## Evidence lives in separate views

The engineer finds the run, compares row counts, searches logs, and decides what to inspect next.

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01 Select the failed run

02 Review cited status, counts, and logs

03 Compare candidate explanations

04 Human verifies before action

**Expected benefit: fewer context switches. The impact study will measure time and evidence selection.**

PRODUCT WORKFLOW

# Operational evidence flows through Exasol, FastAPI, and the dashboard

01 **Pipeline runs and logs** →

02 **Exasol SQL analytics** →

03 **FastAPI rules and candidates** →

04 **Dashboard + optional LLM narrative** →

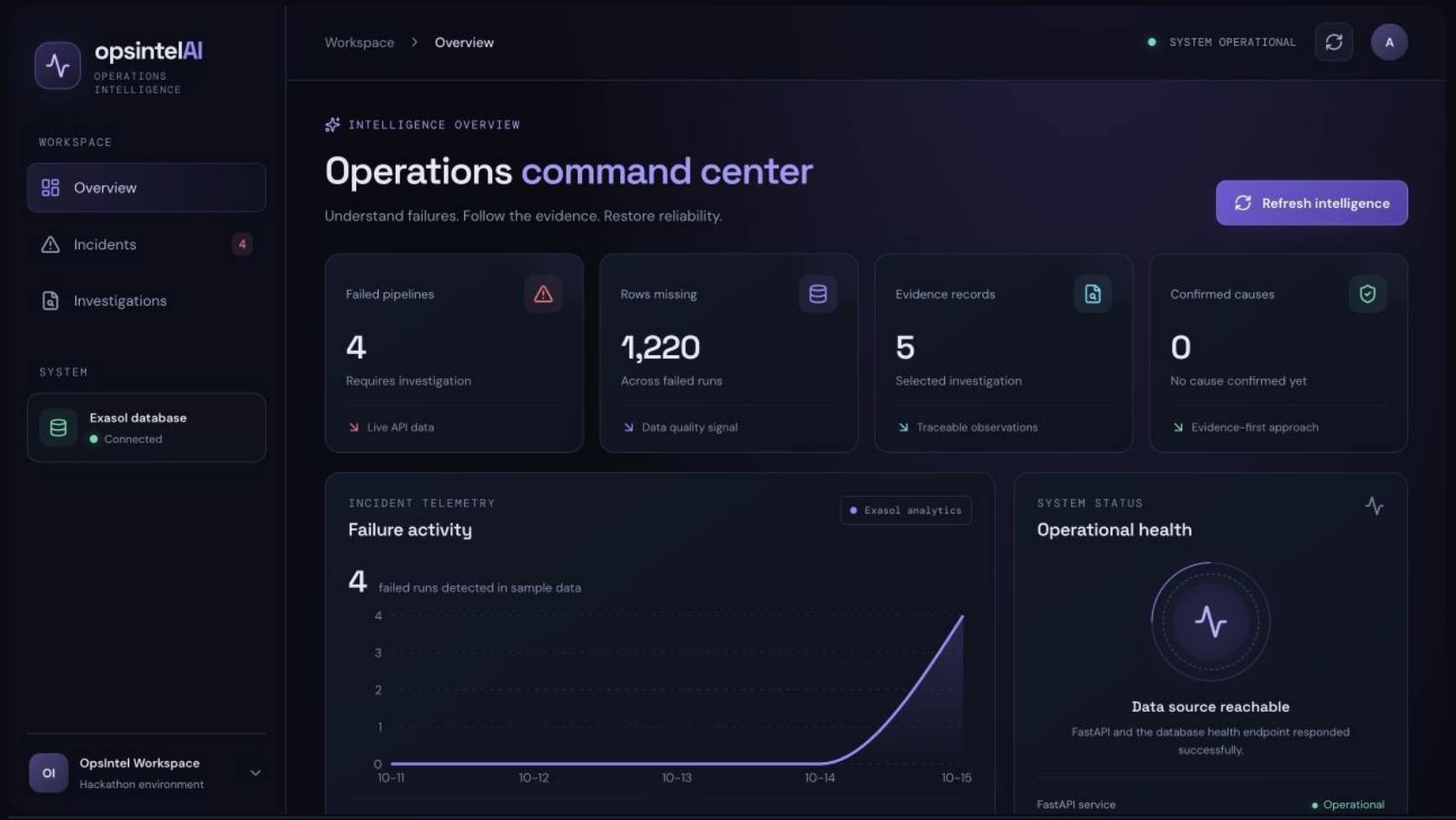
05 **Engineer verifies and confirms**

**Safety boundary**

Rules produce the category and candidates. The optional LLM only summarizes retrieved evidence. Confirmation and repairs stay human-controlled.

LIVE PRODUCT

# The dashboard reads incident telemetry calculated in Exasol




**4**

failed pipeline runs

**1,220**

aggregate missing rows across four demo incidents

✓ Exasol connected
✓ Trend from SQL view
✓ Incidents selectable

INVESTIGATION EXPERIENCE

# Every recommendation points back to source evidence

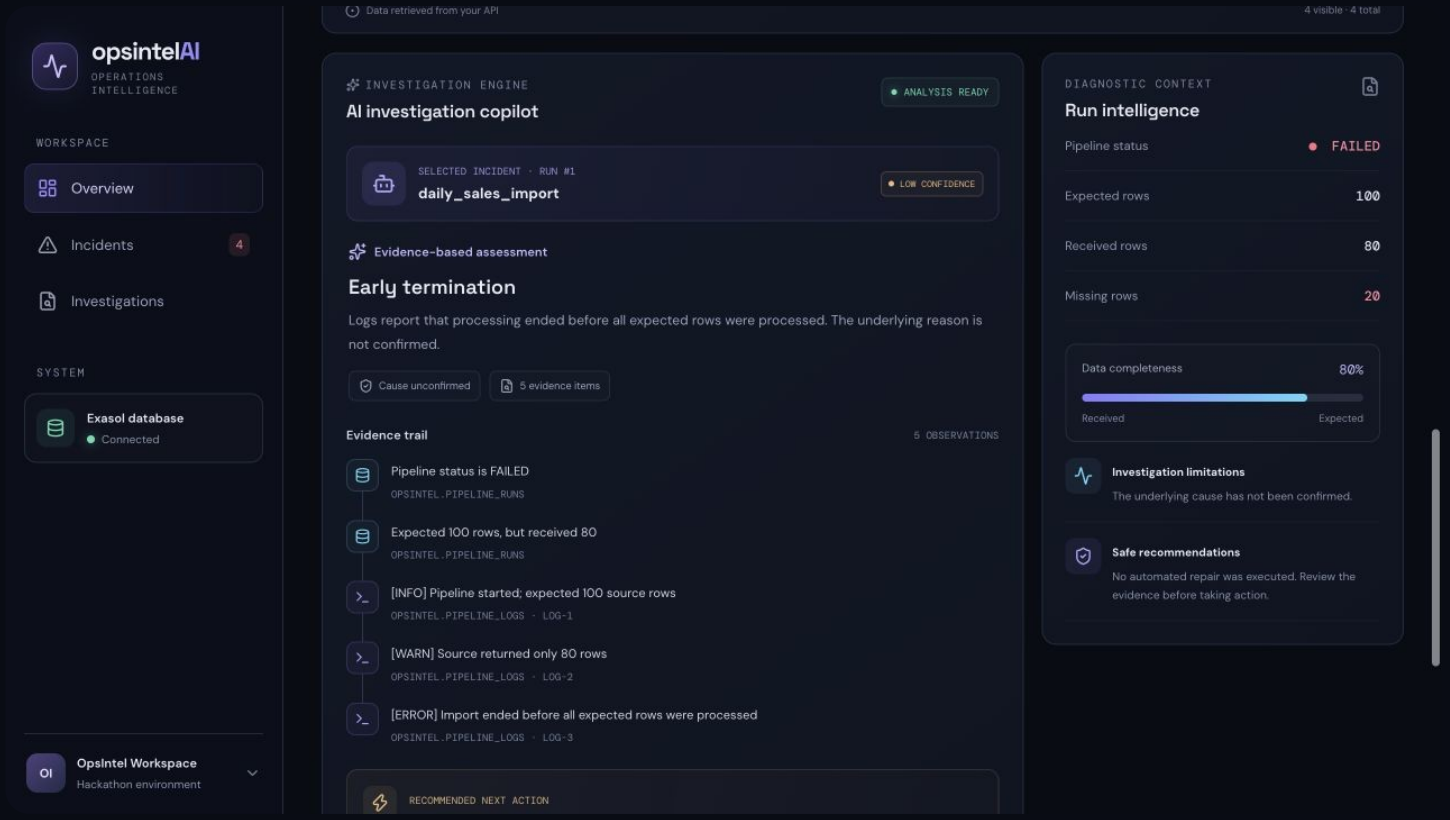
**Selected run**

## daily_sales_import

**OBSERVATION**

Expected: 100 rows
Received: 80 rows
Missing: 20 rows

**EVIDENCE TRAIL**
**PIPELINE_RUNS: status and row counts**
**LOG-1: expected 100 source rows**
**LOG-2: source returned only 80 rows**
**LOG-3: import ended before completion**




**CANDIDATES**

Source shortfall or early termination
Confidence: low

**CONFIRMATION**

None
Human verification required

WHY EXASOL

# Exasol computes incident context before FastAPI responds

## ANALYTICAL FEATURES

01 Previous run status with LAG

02 Rolling actual-row baseline

03 Rolling failure rate

04 Missing rows and completeness

05 FastAPI reads the view directly

### QUERY USED BY /analytics/trend

```
SELECT TO_CHAR(RUN_TIME, 'YYYY-MM-DD') AS RUN_DAY,
       COUNT(*) AS TOTAL_RUNS,
       SUM(CASE WHEN RUN_STATUS = 'FAILED'
         OR EXPECTED_ROWS > ACTUAL_ROWS THEN 1 ELSE 0 END)
         AS INCIDENT_COUNT,
       ROUND(AVG(COMPLETENESS_PCT), 1) AS AVG_COMPLETE_PCT
FROM OPSINTEL.PIPELINE_HEALTH_FEATURES
GROUP BY TO_CHAR(RUN_TIME, 'YYYY-MM-DD')
```

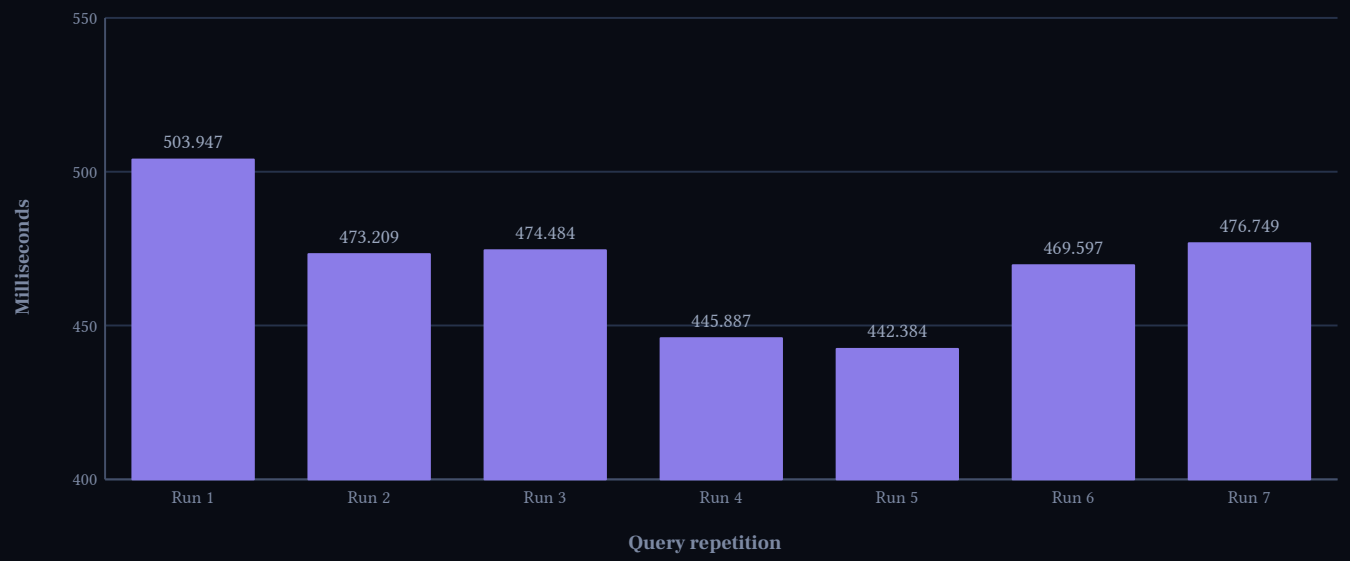
### ACTUAL OUTPUT FROM LOCAL EXASOL

**2026-10-15  5 runs  4 incidents**
**56.0% average completeness  16.0% rolling failure rate**

FastAPI returns these view rows directly. Exasol keeps the window calculations beside the run history, so Python does not rebuild them.

MEASURED EVIDENCE

# 1,000,000 synthetic run rows: 473 ms median and 504 ms p95




**16 / 16**

predefined synthetic challenge cases

**3.025 s**

1,000,000-row import

**473 ms**

median analytical query

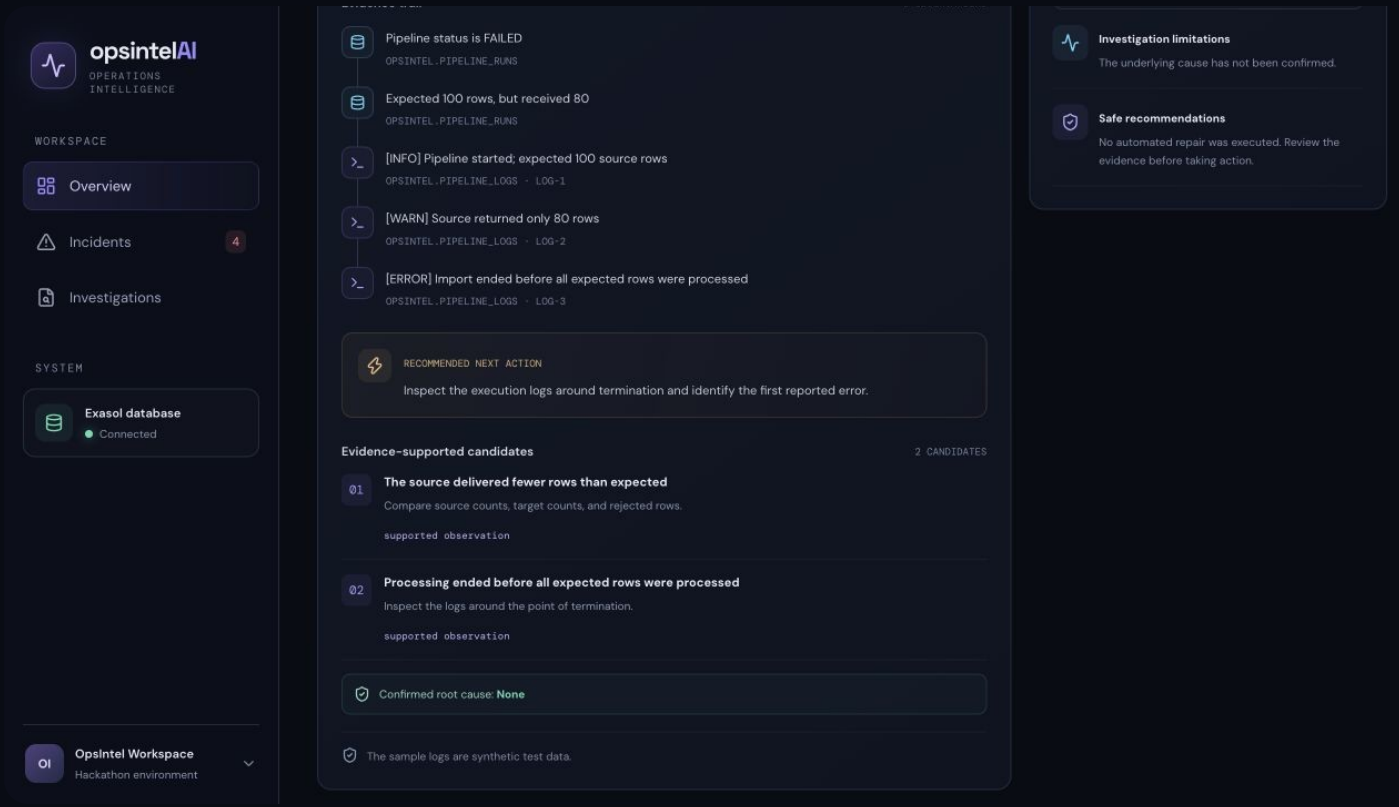
**60/60**

automated tests passed

Records: reports/exasol-benchmark.json. Reproduce with scripts/benchmark_exasol.py, pytest -q, and tests/run_evaluation.py. Local single-user, 7 repeats. No concurrency claim.

TRUST AND SAFETY

# The product is designed to abstain before it overclaims




OBSERVATION

## Source-backed facts

CANDIDATE

## Evidence-supported possibility

CONFIRMATION

## Reserved for human verification

INSUFFICIENT EVIDENCE EXAMPLE
Worker exited with code 71. No diagnostic details. Result: unclassified. Optional summary cannot confirm a cause or run a repair.

VALIDATION ROADMAP

# What is proven, and what must be proven next

## PROVEN NOW

01 Live Exasol-backed demo

02 One-million-row benchmark

03 60 automated tests

04 Optional structured LLM summary

## DEFINED, NOT COMPLETED

01 Reviewer holdout protocol

02 Frozen-commit process

03 Human impact study runner

04 LLM summary quality evaluation

## PRODUCTION WORK

01 Orchestrator connectors

02 Authentication and authorization

03 Concurrent workload testing

04 Audited retention and monitoring

**No independent holdout result or measured productivity improvement is claimed.**

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# Follow the evidence. Investigate faster. Confirm carefully.

github.com/Anafa-thabassum/opsintel-ai

Live demo: one incident, five observations, one safe next action